

Crop Bot - Regional Yield Screening Evidence

Contribution

Samuel Faulkner built and checked a hackathon presentation application for four-region corn-hybrid screening. The work joined data reconciliation, interactive decision support and supplied-model inference replay.

Recorded work

The application exposes parent-pair and location predictions, regional thresholds, review flags and downloadable candidate rows. It uses supplied model checkpoints and a two-seed ensemble, rather than training a new model inside the app.

Recorded outcomes

Using the supplied row-level region labels corrected a regional count mismatch. The curated record reports inference replay differences below 0.001 bushels per acre across all four regions. That agreement measures reproduction of supplied predictions, not accuracy against future yields.

Evaluation discipline

The workflow keeps RMSE, normalized RMSE, R-squared, rank correlation, threshold accuracy, precision, recall and baseline behavior separate. Retrospective evaluation and weak regional performance remain visible rather than being converted into a misleading accuracy headline.

Source and scope

Source: PPP Crop Bot Regional Yield Screening Experience and its linked experience-pool record, reviewed September 21, 2026. These are self-reviewed historical records, not an independent audit. No new model test was executed for this portfolio update. Raw competition data, checkpoints and private workspace material are not included.

Team recognition

Samuel Faulkner placed third in the AI hackathon as a member of team Crop Circuits. Samuel confirmed the team name and result on September 21, 2026 and reports having photographs. The award is included on the CV and portfolio. The proof photograph belongs in supporting documentation, not the CV, and has not yet been attached to this report.

Presentation opportunity

Samuel reports that Crop Circuits is scheduled to present at the AI in Agriculture Symposium on September 21, 2026. The presentation is recorded as scheduled, not completed; the year follows the current hackathon context.